# Is Meta-Learning Effective for Few-Shot Hyperspectral Image Classification?

Li Shen, Yangzhu Wang, Xiaoman Zhang, Huaxin Qiu, Zhenyang Zhang, Chang Nie, and Wei Li

***Abstract*—Recently, there has been a surge of meta-learning-based approaches for the few-shot hyperspectral image classification (FSHSIC) task. Meta-learning leverages prior knowledge to teach a base learner how to adapt quickly to a new few-shot task, which hinges on the consistency of the prior and new tasks to guarantee validity. However, hyperspectral image classification (HSIC) is an environment-dependent task, which means that the hyperspectral features of two objects in the same category can be essentially distinctive in different environments. Consequently, whether meta-learning is a feasible solution for FSHSIC is an imperative problem to investigate, notwithstanding the promising performance shown in previous meta-learning-based approaches. To this end, this work proposes a simple multilayer perceptron (MLP)-based model named SimHSIC for FSHSIC. SimHSIC utilizes only a few labeled samples from the target HSI to train the model rapidly. Surprisingly, SimHSIC outperforms existing meta-learning-based approaches, which are built upon complex 3-D convolutions or transformers and need heavy training processes, in prevailing public benchmarks. On the basis of extensive experiments, we conclude that the relatively better classification performances of meta-learning-based FSHSIC solutions are due mainly to the patching of each HSI pixel with the large surroundings instead of meta-learning. The code is released on https://github.com/OrigamiSL/SimHSIC**

***Index Terms*—Few-shot learning, hyperspectral image classification (HSIC), meta-learning, patch.**

## I. Introduction

COMPARED with RGB images, hyperspectral images (HSIs) display richer object information because they represent a group of narrow spectral bands over a continuous spectral range of the electromagnetic spectrum. Hyperspectral image classification (HSIC) is a long-standing task for distinguishing the objects in a given HSI, and its applications cover a broad array of fields, including but not limited to agriculture [1], mineralogy [2], and biomedicine [3]. Since the labeling of HSI data is difficult and expensive in real-world practice, few-shot hyperspectral image classification (FSHSIC) is the mainstream research direction of HSIC. Given a fully labeled prior HSI as the source data and a few labeled samples in the new target HSI, an FSHSIC task aims to classify the remaining instances in the target HSI.Meta-learning [4], [5], [6], [7] is arguably one of the most successful solutions for few-shot learning because it demonstrates unparalleled performance in a plethora of few-shot learning cases, such as image classification [4], image segmentation [5], object detection [6], and visual tracking [7]. Thus, meta-learning was believed to be a prominent solution [8], [9], [10], [11], [12] for FSHSIC in recent years. However, meta-learning is solely able to recover reasonable solutions for new unseen scenarios if and only if the data from the new tasks are not heavily outside the distribution of training tasks [13]. This means that the advantage of meta-learning is limited and unstable when tackling severe out-of-distribution tasks, wherein the data distributions of the prior and new tasks are extremely distinct. As demonstrated by Setlur et al. [14], the meta-learning method is problematic and heavily relies on model selection under an out-of-distribution setting. Unfortunately, HSIC is such a case [15]. This is because materials often have varying reflectances depending on their angles of illumination and view, sensor characteristics, atmospheric effects, and even biological factors, such as species variation or aging. As a result, the same material may exhibit different spectral features in different images, even if the spectral resolution and wavelength ranges are identical, let alone when the images differ in their spectral coverage and resolution. For example, HSIs of water from different corners of the same lake can exhibit significant distribution discrepancies due to variations in water quality and microbial content. Fig. 1 shows the hyperspectral reflectance spectra of water from different districts in a lake located in Huzhou, Zhejiang, China. In fact, a handful of previous research studies [16], [17], [18], [19], [20] have discovered the distribution shift problem between source and target HSIs. These approaches assume a slight distribution shift and attempt to implement distribution transfer processes to alleviate the problem. However, they are not well-suited for heavy distribution shift problems, and normally, they are merely able to transfer invariant patterns; hence, their approaches fail to be feasible solutions for FSHSIC in real-world practice. In addition, the source and target datasets are acquired via different HSI cameras, leading to variations in the wavelength ranges and the number of spectral bands. Although adjacent spectral bands in hyperspectral imaging exhibit high

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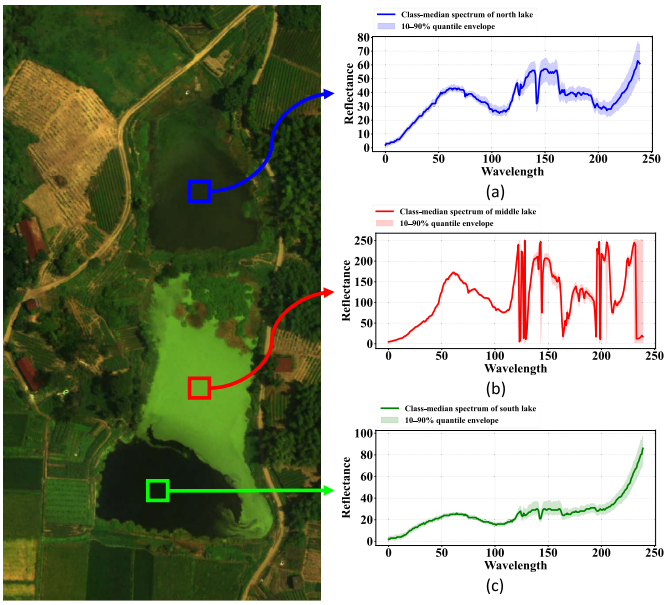



Fig. 1. Reflectance spectra of water samples collected from multiple regions within a lake in Huzhou, Zhejiang, China. Blue, red, and green curves depict the class-median reflectance of three different local districts at each wavelength. The corresponding shaded bands show the p-quantile envelopes, formed from the reflectance distribution's 10% and 90% quantiles at each wavelength. (a)–(c) Samples from the north lake, middle lake, and south lake, respectively.

correlation, the inherent nonlinear relationships [21] between these adjacent bands are often neglected, which means that even if two bands from the source and target datasets are spectrally close, alignment remains challenging. In addition, arbitrary bands can be more related to adjacent bands within a certain range, whereas the relationship with distant bands is usually very weak [22], [23], making the alignment of spectral features between the two datasets particularly difficult, especially in cases where the corresponding bands are spectrally distant. However, previous approaches simply employed linear projections to implement the spectral band alignment of the source and target HSIs. The DCFSL [19] method first applies $1 \times 1$ convolutional mapping layers to both the source and target domains, linearly projecting the original spectral vectors into a unified dimensional space. Similarly, SCFormer [24] uses $1 \times 1$ convolutional mappings to compress spectral bands with different resolutions into a common dimension. However, both methods rely solely on simple linear projection strategies, which are insufficient for achieving precise spectral alignment between the source and target domains. In essence, different spectral bands are deemed homogeneous by these approaches, which is frequently contradictory to the truth. On the basis of the aforementioned analysis, this work poses a disruptive question: Is meta-learning literally effective for FSHSIC?

To answer this question, this work proposes a simple multilayer perceptron (MLP)-based model named SimHSIC as an FSHSIC baseline for comparison with existing state-of-the-art meta-learning-based models. SimHSIC merely implements feature extraction of the spectral bands with one MLP module. In addition, its architecture does not involve any mechanism for spatial feature extraction, and the spatial features are simply averaged at the end of the model, whereas complex spectral–spatial feature extraction is prevalent in existing meta-learning-based methods. Moreover, SimHSIC does not use any source HSI for prior knowledge.

Surprisingly, our empirical results on four prevailing public benchmarks (Indian Pines (IP) [25], Pavia University (UP) [26], Houston 2013 (HT) [27], and Salinas (SA) [28]) show that SimHSIC considerably outperforms existing state-of-the-art meta-learning-based models in most FSHSIC cases in terms of accuracy, efficiency, and robustness. A brief comparison is shown in Fig. 2. Furthermore, in contrast to the claims in existing works involving meta-learning-based FSHSIC, most of them rely on the technique of representing each HSI pixel by its surrounding large patch to achieve relatively better classification performance. We call this a "large patching technique" (LPT) for brevity in the following sections of this work. Indeed, LPT has been proven to be effective in various realms, including FSHSIC. Moreover, meta-learning has been shown to contribute little to performance enhancement according to the limited and unstable gains with meta-learning in the following experiments, while LPT brings consistent and immense benefits to the FSHSIC approaches. The main contributions of this work can be summarized into three points as follows.

1) This work reveals that the performance gains of existing meta-learning-based FSHSIC methods are attributed primarily to the LPT rather than the meta-learning paradigm.
2) This work introduces a simple yet effective MLP-based FSHSIC baseline, which achieves promising classification performance without prior knowledge and with minimal training cost.
3) Extensive experiments validate that SimHSIC achieves state-of-the-art performance compared with meta-learning-based models on four prevailing public benchmarks with peerless accuracy, efficiency, and robustness.

The remainder of the article is organized as follows. Section II briefly introduces related works. Section III defines the FSHSIC. Section IV describes the proposed methods in detail. Section V presents the experiments and analyzes the classification results. Section VI presents the conclusion and discusses the limitations of this work.

## II. Related Works

### A. Few-Shot Hyperspectral Image Classification

The objective of FSHSIC is to develop an effective and highly generalizable HSIC classification model with very few labeled training data. Owing to the success of meta-learning in a plethora of scenarios requiring few-shot learning settings [4], [5], [6], [7], it is also widely used as a paradigm for FSHSIC [19], [29], [30], [31]. For example, BMFSC [32] is a representative optimization-based meta-learning method that employs a Bayesian bilevel framework, where inner- and outer-loop updates are used to infer task-specific variables and update shared parameters for hyperspectral few-shot adaptation. Moreover, SGPL [33] follows a metric-based paradigm, constructing a semantically guided prototype space and employing a relation function to

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compare query and support samples for cross-task generalization. In addition, HCFSL-NAS [34] adopts a model-based meta-learning strategy, where task adaptability is achieved via task-aware architecture search, enabling the model to dynamically select suitable subnetworks conditioned on the support set. However, the discussion of whether meta-learning literally conforms to FSHSIC scenarios is not clear. In contrast, other researchers realized that natural and nonnegligible distribution shifts may exist between the source and target HSIs so that meta-learning cannot be applied individually [35], [36]. To address this challenge, various methods have been proposed to align the source and target domains explicitly. DCFSL [19] employs $1 \times 1$ convolutions to linearly project features from both source and target domains into a unified space. Although the projection functions are learned separately, the network structures are identical for both domains, lacking an explicit mechanism for domain-specific adaptation. SCFormer [28] similarly uses $1 \times 1$ convolutions to compress spectral bands with different resolutions, applying the same compression structure to both domains. This design implicitly assumes that the spectral features become aligned after projection, ignoring the inherent spectral variability between domains. TEFSL [12] introduces spectral–spatial attention and adaptive normalization to guide feature alignment. Although it incorporates nonlinear components, such as attention and graph-based refinement, the overall architecture remains shared across domains and lacks explicit mechanisms to model domain-specific discrepancies. FDFSL [11] proposes a feature disentanglement framework that orthogonally decomposes source and target features into domain-shared and domain-specific components. While structurally more sophisticated, it still relies on a shared projection space. Nevertheless, all these methods fall short in addressing the strong distribution shift between source and target domains in HSIC. Despite efforts to improve the meta-learning-based FSHSIC, it remains doubtful that meta-learning is rigorously effective for FSHSIC because of the conflict between its requirement of slight out-of-distribution and severe distribution shifts between heterogeneous HSIs.

### B. Patching Technique

Patching is a well-known technique in numerous AI fields because it improves model efficiency and enlarges the receptive field. Vision transformer (ViT) [37] is a widely recognized model in computer vision that employs a patching technique to divide an image into fixed-size patches. Numerous patch-based vision models, such as Swin Transformer [38], DeiT [39], CaiT [40], and T2T-ViT [41], have evolved from ViT owing to its success. Moreover, in time series analysis tasks, patching has been shown to be an effective operator for processing temporal information [42], which can be easily transferred to nearly all types of models, such as transformer-based DSformer [43], CNN-based TSLANet [44], and RNN-based WITRAN [45]. In addition, in the acoustic domain, patching has also been used to simultaneously process the temporal and spectral information of audio signals, which facilitates structured time–frequency representation learning and can be effectively integrated into various transformer-based models, such as Audio-MAE [46], CED [47], and EAT [48]. Classifying each HSI pixel by the surrounding patch instead of merely by itself has also been shown to be more effective [49], [50], [51], [52]. However, this work empirically demonstrates that the patching technique is much more crucial than previously thought for FSHSIC.

## III. Problem Statement

The FSHSIC problem is defined as classifying HSI pixels with a limited number of labeled examples per class. Formally, the task is structured as an $N$-way $K$-shot classification problem, and the training dataset is organized as follows: $\mathcal{D}_{\text{train}} = \{(x_a, y_a)\}_{a=1}^{I}$, where $I = K \times N$ denotes the number of labeled samples, with exactly $K$ examples provided for each of the $N$ classes. Specifically, each training sample $x_a^i \in \mathbb{R}^{P\times P\times B}$ is a spatial–spectral patch extracted from an HSI $\mathcal{X}$. It consists of a square region of size $P \times P$ centered on a single pixel and includes $B$ spectral bands per pixel. Note that only the center pixel is labeled, and the surrounding pixels are assumed to belong to the same class as the center pixel. Note that only the center pixel is labeled, and the surrounding pixels are assumed to fall into the same class as the center pixel. The corresponding label $y_a \in \{1, 2, \ldots, N\}$ indicates the class of the center pixel in the patch.

## IV. Methodology

### A. Model Architecture

As illustrated in Fig. 3, SimHSIC comprises an embedding layer, a backbone, and a head. Instead of the group convolutions commonly used in previous approaches [8], [9], [11], the embedding layer of SimHSIC adopts a simple linear projection layer followed by batch normalization and a GELU [53] activation function. The algorithmic process is formulated as follows:

$$\boldsymbol{X}_a = \text{GELU}\left(\text{BN}\left(\boldsymbol{X}\boldsymbol{W}_1\right)\right) \tag{1}$$

where $\boldsymbol{X} \in \mathbb{R}^{P\times P\times B}$ is the input spatial–spectral patch and where $\boldsymbol{X}_a \in \mathbb{R}^{P\times P\times D}$ is the embedded representation. $\boldsymbol{W}_1 \in \mathbb{R}^{B\times D}$ is a linear projection matrix that maps the input spectral channels to the hidden feature space with a dimensionality of $D$. BN$(\cdot)$ denotes the batch normalization.

The embedded output $\boldsymbol{X}_a$ then propagates through an MLP-based backbone as follows:

$$\boldsymbol{X}_b = \text{LN}\left(\text{LN}\left(\boldsymbol{X}_a\right) + \text{GELU}\left(\text{LN}\left(\boldsymbol{X}_a\right)\boldsymbol{W}_2\right)\boldsymbol{W}_3\right) \tag{2}$$

$$\boldsymbol{X}_c = \text{GELU}\left(\text{BN}\left(\boldsymbol{X}_b\boldsymbol{W}_4 + \boldsymbol{X}_a\right)\right) \tag{3}$$

where $\boldsymbol{X}_c \in \mathbb{R}^{P\times P\times D}$ is the output of the backbone. $\boldsymbol{W}_2, \boldsymbol{W}_3, \boldsymbol{W}_4 \in \mathbb{R}^{D\times D}$ are linear projection matrices used in the MLP block to transform and refine intermediate features with residual connections. LN$(\cdot)$ denotes layer normalization.

The head of SimHSIC consists of two parts. The first part is designed to learn representations of the input. Its process is as follows:

$$\boldsymbol{Y}_a = \text{LN}\left(\text{AvgPool}\left(\boldsymbol{X}_c\right)\right)\boldsymbol{W}_5\boldsymbol{W}_6 \tag{4}$$

where $\boldsymbol{Y}_a \in \mathbb{R}^{D'}$ is the ultimate representation of $\boldsymbol{X}$ in the first head and $D'$ is the dimension of the output latent feature

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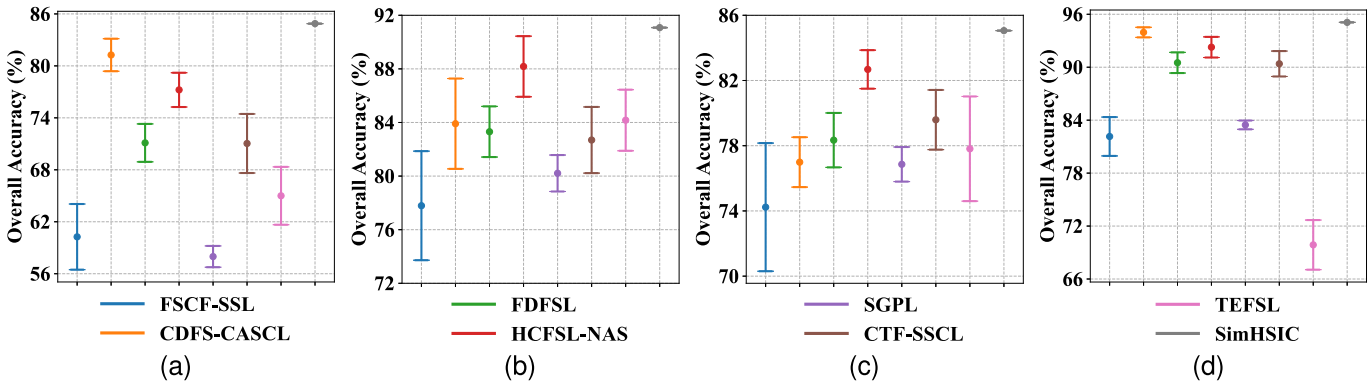



Fig. 2. Brief comparison between SimHSIC and several state-of-the-art meta-learning-based models on four public benchmarks in terms of model performance (overall accuracy and the corresponding error bar). SimHSIC demonstrates superior results. (a) IP. (b) UP. (c) HT. (d) SA.

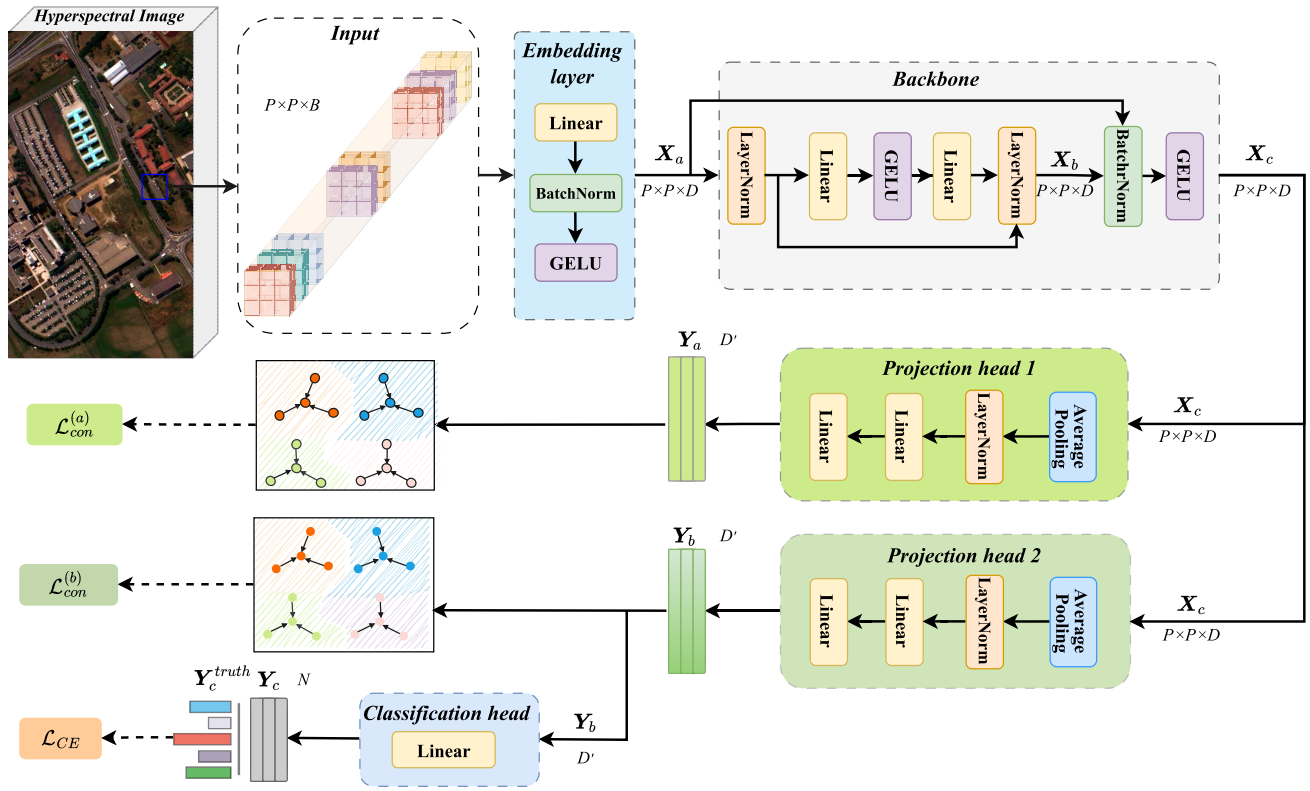



Fig. 3. Overview of the proposed SimHSIC model. The architecture includes an embedding layer for spectral feature projection, a backbone for representation learning, and a dual-branch head. $\boldsymbol{Y}_a$ and $\boldsymbol{Y}_b$ are both used for contrastive learning but $\boldsymbol{Y}_b$ further supports classification via a linear head to produce $\boldsymbol{Y}_c$. The model is jointly optimized by three losses: $\mathcal{L}_{\text{con}}^{(a)}$, $\mathcal{L}_{\text{con}}^{(b)}$, and $\mathcal{L}_{\text{CE}}$.

space. AvgPool(·) averages across the spatial dimension. $\boldsymbol{W}_5 \in \mathbb{R}^{D\times 2\text{-D}}$ and $\boldsymbol{W}_6 \in \mathbb{R}^{2-D\times D'}$ are two linear projection matrices.

The second part applies another projection head—identical in structure to the first—to generate a latent representation, denoted as $Y_b$. This representation is further used to compute the direct classification output $Y_c$. The computation process is defined as follows:

$$\boldsymbol{Y}_b = \text{LN}\left(\text{AvgPool}\left(\boldsymbol{X}_c\right)\right)\boldsymbol{W}_7\boldsymbol{W}_8 \tag{5}$$

$$\boldsymbol{Y}_c = \boldsymbol{Y}_b\boldsymbol{W}_9 \tag{6}$$

where $\boldsymbol{Y}_b \in \mathbb{R}^{D'}$ is the latent representation produced by the second head, which shares the same dimensionality $D'$ as $\boldsymbol{Y}_a$. $\boldsymbol{Y}_c \in \mathbb{R}^N$ is the classification output indicating the confidence scores over $N$ classes. Similarly, $\boldsymbol{W}_7 \in \mathbb{R}^{D\times 2\text{-D}}$, $\boldsymbol{W}_8 \in \mathbb{R}^{2-D\times D'}$, and $\boldsymbol{W}_9 \in \mathbb{R}^{D'\times N}$ are the weight matrices for the linear projection layers.

## B. Training and Inference

Following CTF-SSCL [10], two training objectives are constructed for SimHSIC, as illustrated in Fig. 3. The first training objective is to teach the model to classify input directly via the classification output $\boldsymbol{Y}_c$. To this end, the cross-entropy loss $\mathcal{L}_{\text{CE}}$ between $\boldsymbol{Y}_c$ and the ground-truth label distribution $\boldsymbol{Y}_c^{\text{truth}}$ is adopted

$$\mathcal{L}_{\text{CE}} = -\frac{1}{n}\sum_{i=1}^{n}\boldsymbol{Y}_{i,c}^{\text{truth}}\cdot\log\left(\boldsymbol{Y}_{i,c}\right) \tag{7}$$

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TABLE I
DETAILED INFORMATION REGARDING FOUR DATASETS

| Dataset | Resolution (m) | Spatial Size | Effective Band | Classes | Location | Sensor |
|---|---|---|---|---|---|---|
| IP | 20 | $145 \times 145$ | 200 (from 224) | 16 | Northwestern Indiana, USA | AVIRIS |
| UP | 1.3 | $610 \times 340$ | 103 (from 115) | 9 | Pavia, Italy | ROSIS |
| HT | 2.5 | $340 \times 1905$ | 144 | 15 | Houston, Texas, USA | CASI |
| SA | 3.7 | $512 \times 217$ | 204 (from 224) | 16 | Salinas Valley, California, USA | AVIRIS |

where $n$ is the number of samples. In addition, the second training objective is to teach the model to distinguish the representations of pixels belonging to different classes via contrastive learning on the basis of the feature outputs $\boldsymbol{Y}_a$ and $\boldsymbol{Y}_b$. Suppose that the model input consists of a set of samples $\widetilde{\boldsymbol{X}} = \{\boldsymbol{X}_1, \boldsymbol{X}_2, \ldots, \boldsymbol{X}_n\}$ and that the corresponding learned representations are $\widetilde{\boldsymbol{Y}}_a = \{\boldsymbol{Y}_{1,a}, \boldsymbol{Y}_{2,a}, \ldots, \boldsymbol{Y}_{n,a}\}$ and $\widetilde{\boldsymbol{Y}}_b = \{\boldsymbol{Y}_{1,b}, \boldsymbol{Y}_{2,b}, \ldots, \boldsymbol{Y}_{n,b}\}$. The contrastive learning loss $\mathcal{L}_{\text{con}}^{(a)}$ is formulated as

$$\mathcal{L}_{\text{con}}^{(a)} = -\frac{1}{n}\sum_{i=1}^{n} \log \frac{\sum \exp\left(Y_{i,a}^{+}\right)}{\sum \exp\left(Y_{i,a}^{+}\right) + \sum \exp\left(Y_{i,a}^{-}\right)} \tag{8}$$

where $Y_i^+$ and $Y_i^-$ represent the similarities with positive and negative samples, respectively. Let $Y_{t,a}$ denote the feature embedding obtained by encoding the input sample, where $t \in \{i, j, k\}$ and $i, j, k \in \{1, 2, \ldots, n\}$ with $i \neq j \neq k$. The class label of $Y_{t,a}$ is denoted as $c_t$. Here, $Y_{i,a}$ is the anchor, $Y_{j,a}$ serves as the positive sample if $c_i = c_j$, and $Y_{k,a}$ is taken as the negative sample if $c_i \neq c_k$. The specific similarity formulations are defined as follows:

$$\begin{aligned} Y_{i,a}^{+} &= \text{sim}\left(\boldsymbol{Y}_{i,a}, \boldsymbol{Y}_{j,a}\right), \quad \text{if } c_i = c_j \\ Y_{i,a}^{-} &= \text{sim}\left(\boldsymbol{Y}_{i,a}, \boldsymbol{Y}_{k,a}\right), \quad \text{if } c_i \neq c_k \end{aligned} \tag{9}$$

$$\text{sim}(\boldsymbol{u}, \boldsymbol{v}) = \frac{\boldsymbol{u}^{\top}\boldsymbol{v}}{\|\boldsymbol{u}\| \cdot \|\boldsymbol{v}\|} \tag{10}$$

where $\text{sim}(\cdot, \cdot)$ denotes the cosine similarity. Similarly, the contrastive learning loss $\mathcal{L}_{\text{con}}^{(b)}$ is defined in the same manner based on $\widetilde{\boldsymbol{Y}}_b$.

On the basis of the aforementioned three losses, the complete loss function of SimHSIC is described as follows:

$$\mathcal{L}_{\text{total}} = \lambda_1 \mathcal{L}_{\text{CE}} + \lambda_2 \mathcal{L}_{\text{con}}^{(a)} + \lambda_3 \mathcal{L}_{\text{con}}^{(b)} \tag{11}$$

where $\lambda_1, \lambda_2$, and $\lambda_3$ are balancing hyperparameters that control the contribution of each loss term. This formulation ensures that the model jointly learns to classify and produce semantically meaningful representations. In the inference phase, on the basis of the model outputs, the final classification result is computed for SimHSIC. Specifically, only the classification output $\boldsymbol{Y}_c$ is used to determine the predicted label for each input sample. The prediction vector $\hat{\boldsymbol{Y}}_c \in \{0, 1\}^N$ is a one-hot encoded vector, where the nonzero entry corresponds to the predicted class. A SoftMax function is first applied to convert the logits into a probability distribution over all classes

$$\text{SoftMax}\left(\boldsymbol{Y}_c(j)\right) = \frac{\exp\left(\boldsymbol{Y}_c(j)\right)}{\sum_{j'=1}^{N} \exp\left(\boldsymbol{Y}_c(j')\right)}, \quad j \in \{1, \ldots, N\}. \tag{12}$$

Then, the predicted class index $j^*$ is determined by selecting the one with the highest probability

$$j^* = \arg\max_j \left(\text{SoftMax}\left(\boldsymbol{Y}_c(j)\right)\right). \tag{13}$$

Formally, the one-hot prediction vector $\hat{\boldsymbol{Y}}_c$ is defined as

$$\hat{\boldsymbol{Y}}_c(j) = \begin{cases} 1, & \text{if } j = j^* \\ 0, & \text{otherwise.} \end{cases} \tag{14}$$

## V. EXPERIMENTS

### A. Benchmarks and Baselines

1 *Benchmarks*: Extensive experiments are conducted on four commonly used benchmarks: IP, UP, HT, and SA. These datasets cover diverse geographic locations and vary in spatial resolution, spectral band count, and scene composition. Table I summarizes the key properties of each dataset, including spatial size, spectral dimensionality, ground sampling distance, and sensor type. Table II lists the land-cover categories and the corresponding number of labeled samples per class. These datasets address a broad spectrum of challenges, ranging from agricultural scenes to urban scenes.

2 *Baselines*: Seven representative baseline methods, along with the proposed SimHSIC, are employed for comparative evaluation in FSHSIC. All the baselines are designed on the basis of meta-learning strategies while differing in the types of spatial and spectral feature extractors they adopt. Their architectural characteristics are summarized in Table III. In contrast to these meta-learning-based baselines with complicated feature extractors, SimHSIC employs an MLP-based feature extractor, which offers structural simplicity and competitive effectiveness, without the assistance of any meta-knowledge or source HSI.

### B. Implementation Details

1 *Set Up*: For fair comparisons, the spatial patch size is fixed at $9 \times 9$ across all the experiments, regardless of

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TABLE II
DETAILED NUMBERS OF THE SAMPLES FOR EACH CLASS IN THE FOUR DATASETS

| IP | | UP | | HT | | SA | |
|---|---|---|---|---|---|---|---|
| Land-covers | Samples | Land-covers | Samples | Land-covers | Samples | Land-covers | Samples |
| Alfalfa | 46 | Asphalt | 6631 | Healthy Grass | 1251 | Brocoli green weeds 1 | 2009 |
| Corn-notill | 1428 | Meadows | 18649 | Stressed Grass | 1254 | Brocoli green weeds 2 | 3726 |
| Corn-mintill | 830 | Gravel | 2099 | Synthetic Grass | 697 | Fallow | 1976 |
| Corn | 237 | Trees | 3064 | Trees | 1244 | Fallow rough plow | 1394 |
| Grass-pasture | 483 | Painted metal sheets | 1345 | Soil | 1242 | Fallow smooth | 2678 |
| Grass-trees | 730 | Bare Soil | 5029 | Water | 325 | Stubble | 3959 |
| Grass-pasture-mowed | 28 | Bitumen | 1330 | Residential | 1268 | Celery | 3579 |
| Hay-windrowed | 478 | Self-Blocking Bricks | 3682 | Commercial | 1244 | Grapes untrained | 11271 |
| Oats | 20 | Shadows | 947 | Road | 1252 | Soil vinyard develop | 6203 |
| Soybean-notill | 972 | | | Highway | 1227 | Corn senesced green weeds | 3278 |
| Soybean-mintill | 2455 | | | Railway | 1235 | Lettuce romaine 4wk | 1068 |
| Soybean-clean | 593 | | | Parking Lot1 | 1233 | Lettuce romaine 5wk | 1927 |
| Wheat | 205 | | | Parking Lot2 | 469 | Lettuce romaine 6wk | 916 |
| Woods | 1265 | | | Tennis Court | 428 | Lettuce romaine 7wk | 1070 |
| Buildings-Grass-Trees-Drives | 386 | | | Running Track | 660 | Vinyard untrained | 7268 |
| Stone-Steel-Towers | 93 | | | | | Vinyard vertical trellis | 1807 |
| **Total** | **10249** | **Total** | **42776** | **Total** | **15029** | **Total** | **54129** |

TABLE III
PROPERTIES OF COMPARED METHODS

| Method | FSCF-SSL [8] | CDFS-CASCL [9] | FDFSL [11] | HCFSL-NAS [34] | SGPL [33] | CTF-SSCL [10] | TEFSL [12] | SimHSIC |
|---|---|---|---|---|---|---|---|---|
| Meta-learning | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | × |
| Spatial Feature Extractor | 2D-CNN | 2D-CNN | 3D-CNN | 3D-CNN | 3D-CNN | CNN-Transformer | 3D-CNN | × |
| Spectral Feature Extractor | 1D-CNN | 1D-CNN | 3D-CNN | 3D-CNN | 3D-CNN | CNN-Transformer | 3D-CNN | MLP |

the original configurations used by the baseline methods (e.g., 33 × 33, 28 × 28, or 7 × 7). The number of spectral bands $B$ depends on the dataset and is set to 200 (IP), 103 (UP), 144 (HT), or 204 (SA). Five labeled samples per class are randomly selected for training. To ensure robustness, each experiment is repeated ten times with different random seeds, and the means and standard deviations of all the metrics are reported. The best results are highlighted in **bold** font, whereas the second-best results are highlighted in *underlined and italic* font.

2 *Network Hyperparameters*: The hidden dimensionality $D$ and $D'$ are fixed at 128 and 32, respectively. The two heads share the same structure but operate independently. One produces classification logits for the cross-entropy loss, and the other outputs representations for two contrastive losses. The overall loss is a weighted sum of these three terms, with weights $\lambda_1 = 1.0$, $\lambda_2 = 0.01$, and $\lambda_3 = 0.01$.

3 *Training Strategy*: The model is trained via the Adam optimizer with an initial learning rate of $10^{-3}$ ($10^{-4}$ for UP). The batch sizes are set to 16 (IP), 36 (UP), 32 (HT), and 30 (SA) according to the number of classes. The maximum number of training epochs is 200, and the dropout rate is 0.1.

4 *Evaluation Metrics*: The evaluation metrics include overall accuracy (OA), average accuracy (AA), and the kappa coefficient (Kappa), which are widely adopted in HSIC tasks.

5 *Other Details*: All the experiments are conducted on a workstation equipped with a 13th Generation Intel Core i9-13900HX CPU and an NVIDIA GeForce RTX 4090 laptop GPU with 16 GB of memory. The implementation is based on Python 3.8.18 and PyTorch 1.11.0 with CUDA 11.3 support.

### C. Quantitative Results

The classification performances of SimHSIC and seven meta-learning-based methods are summarized in Table IV (IP), Table V (UP), Table VI (HT), and Table VII (SA). The following observations can be made.

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TABLE IV
CLASSIFICATION PERFORMANCE (%) ON THE IP DATASET

| Class | FSCF-SSL [8] | CDFS-CASCL [9] | FDFSL [11] | HCFSL-NAS [34] | SGPL [33] | CTF-SSCL [10] | TEFSL [12] | SimHSIC |
|---|---|---|---|---|---|---|---|---|
| 1 | 95.61±5.09 | *99.76±0.73* | 96.59±2.72 | **100.00±0.00** | 96.09±0.22 | 98.54±1.62 | 90.00±13.94 | 66.73±1.84 |
| 2 | 40.08±12.22 | **75.00±5.71** | 55.62±9.97 | *60.01±11.74* | 38.60±0.21 | 57.96±13.99 | 45.62±11.37 | 58.35±1.67 |
| 3 | 46.58±12.13 | *71.42±11.92* | 64.05±7.26 | 61.99±5.13 | 33.57±0.42 | 56.07±8.98 | 49.98±9.89 | **90.51±2.40** |
| 4 | 60.39±19.32 | **98.02±2.54** | 87.07±7.78 | *93.88±6.32* | 69.03±0.28 | 88.19±11.04 | 77.41±11.27 | 91.22±4.19 |
| 5 | 68.22±9.44 | 78.89±9.94 | 75.17±12.50 | **81.90±6.94** | *80.68±0.10* | 77.07±7.54 | 74.27±9.17 | 80.59±1.80 |
| 6 | 75.83±9.69 | 94.81±3.15 | 83.52±10.22 | *95.16±2.68* | 66.11±0.27 | 89.48±5.44 | 86.45±6.05 | **96.28±0.41** |
| 7 | 94.35±5.52 | **100.00±0.00** | *99.57±1.30* | **100.00±0.00** | **100.00±0.00** | **100.00±0.00** | **100.00±0.00** | **100.00±0.00** |
| 8 | 82.71±20.57 | *99.41±1.27* | 83.11±14.36 | 96.72±3.21 | 94.00±0.06 | 90.13±12.93 | 84.80±13.99 | **99.50±1.17** |
| 9 | **100.00±0.00** | **100.00±0.00** | **100.00±0.00** | *99.33±2.00* | **100.00±0.00** | 98.00±4.27 | **100.00±0.00** | **100.00±0.00** |
| 10 | 44.68±20.43 | *80.91±12.16* | 66.10±13.21 | 79.17±7.16 | 58.26±0.95 | 67.22±11.79 | 63.48±10.62 | **94.36±0.90** |
| 11 | 60.77±21.28 | *72.85±8.81* | 64.51±6.05 | 72.37±9.75 | 44.77±0.27 | 64.68±11.96 | 56.50±17.80 | **81.03±2.31** |
| 12 | 38.32±9.01 | *81.56±9.77* | 59.29±8.11 | 53.62±16.90 | 38.62±0.14 | 52.33±10.22 | 45.39±13.42 | **88.10±2.87** |
| 13 | 98.85±1.38 | **99.40±1.09** | 98.70±1.35 | 98.00±1.64 | 97.71±0.02 | 99.10±1.58 | 96.75±5.10 | *99.37±0.31* |
| 14 | 84.65±10.61 | 87.62±10.16 | 88.60±5.47 | *95.78±6.32* | 91.83±0.03 | 84.93±9.66 | 87.06±7.12 | **97.35±0.98** |
| 15 | 51.52±15.47 | **86.85±13.11** | 80.21±11.64 | 81.60±8.55 | 55.13±0.65 | *82.07±10.38* | 70.47±10.96 | 76.45±6.14 |
| 16 | 95.11±5.95 | 97.27±4.64 | 98.18±1.98 | 95.91±9.14 | 71.18±0.46 | *99.55±0.91* | 99.54±1.04 | **100.00±0.00** |
| OA | 60.26±3.79 | *81.26±1.88* | 71.11±2.19 | 77.23±1.98 | 57.98±1.23 | 71.04±3.41 | 65.00±3.34 | **84.88±0.31** |
| AA | 71.10±2.74 | **88.99±1.52** | 81.27±1.84 | 85.34±1.25 | 56.82±1.71 | 81.58±2.44 | 76.73±1.31 | *88.74±0.41* |
| Kappa | 54.88±3.87 | *78.86±2.09* | 67.55±2.44 | 74.17±2.22 | 52.93±1.30 | 67.35±3.78 | 60.61±3.43 | **82.91±0.35** |

TABLE V
CLASSIFICATION PERFORMANCE (%) ON THE UP DATASET

| Class | FSCF-SSL [8] | CDFS-CASCL [9] | FDFSL [11] | HCFSL-NAS [34] | SGPL [33] | CTF-SSCL [10] | TEFSL [12] | SimHSIC |
|---|---|---|---|---|---|---|---|---|
| 1 | 76.07±12.71 | **91.46±7.38** | 86.07±4.22 | *88.85±4.21* | 72.71±0.06 | 88.06±6.42 | 76.52±12.39 | 87.19±2.40 |
| 2 | 80.72±7.40 | 73.57±8.45 | 83.09±8.28 | 89.24±6.05 | *92.19±0.11* | 79.56±7.87 | 90.84±6.32 | **98.21±0.78** |
| 3 | 59.54±11.67 | **84.15±11.34** | 65.16±6.85 | *82.31±10.50* | 61.56±0.18 | 74.89±10.99 | 71.39±9.33 | 79.09±6.77 |
| 4 | *92.86±5.08* | 90.42±4.98 | 89.29±5.67 | 90.45±4.54 | 93.95±0.02 | 90.08±5.16 | **95.49±1.48** | 90.06±2.06 |
| 5 | 99.46±1.02 | 99.75±0.40 | 99.40±0.61 | 99.78±0.42 | 99.98±0.00 | **99.82±0.24** | 99.50±1.06 | *99.81±0.09* |
| 6 | 61.05±14.84 | **93.82±6.05** | 75.11±12.61 | *86.71±7.73* | 48.23±0.22 | 73.54±10.54 | 67.73±13.53 | 71.65±5.52 |
| 7 | 73.64±13.73 | **98.62±1.52** | 88.66±7.90 | 90.03±5.67 | 84.94±0.04 | 91.38±9.57 | 85.50±5.87 | *94.02±1.37* |
| 8 | 74.65±9.34 | *89.37±7.08* | 85.31±9.52 | 77.33±12.03 | 61.94±0.15 | 86.63±3.72 | 74.53±12.53 | **90.79±5.08** |
| 9 | **99.98±0.06** | 96.28±3.71 | 94.62±3.33 | 99.16±1.39 | 99.86±0.00 | 96.94±2.63 | *99.35±0.74* | 98.52±0.52 |
| OA | 77.79±4.07 | 83.91±3.37 | 83.31±1.89 | *88.18±2.26* | 80.21±1.36 | 82.69±2.47 | 84.17±2.28 | **91.08±0.90** |
| AA | 79.77±2.48 | **90.83±1.47** | 85.19±2.23 | 89.32±2.64 | 76.99±1.31 | 86.77±2.06 | 84.54±1.27 | *89.93±0.76* |
| kappa | 71.29±4.91 | 79.72±3.95 | 78.34±2.16 | *84.58±2.87* | 73.67±1.69 | 77.79±2.83 | 79.22±2.87 | **88.15±1.21** |

1) SimHSIC yields average improvements of 19.89%, 4.73%, 9.33%, 4.39%, 20.30%, 8.84%, and 18.87% in comparison with FSCF-SSL, CDFS-CASCL, FDFSL, HCFSL-NAS, SGPL, CTF-SSCL, and TEFSL, respectively, which demonstrates the state-of-the-art performance of SimHSIC.
2) In addition, a very low standard deviation of SimHSIC is observed across repeated trials, indicating that SimHSIC provides excellent generalizability across diverse remote sensing scenarios.
3) SimHSIC achieves competitive results without meta-learning, revealing that the benefits of meta-learning

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TABLE VI
CLASSIFICATION PERFORMANCE (%) ON THE HT DATASET

| Class | FSCF-SSL [8] | CDFS-CASCL [9] | FDFSL [11] | HCFSL-NAS [34] | SGPL [33] | CTF-SSCL [10] | TEFSL [12] | SimHSIC |
|---|---|---|---|---|---|---|---|---|
| 1 | 87.29±7.32 | 79.32±5.94 | 87.35±5.41 | 88.41±7.88 | 84.12±0.00 | *89.48±7.02* | **89.58±6.48** | 85.93±0.35 |
| 2 | 81.85±7.13 | 77.27±6.57 | 86.93±5.70 | *90.61±6.90* | 83.15±0.09 | 84.48±8.00 | 82.30±12.37 | **97.90±0.95** |
| 3 | 96.33±4.22 | *99.19±1.15* | 96.24±2.92 | 97.44±2.58 | 94.23±0.17 | 99.06±0.91 | 97.12±1.76 | **100.00±0.00** |
| 4 | 91.65±2.43 | 90.18±1.95 | 90.70±2.57 | **95.33±2.74** | 95.60±0.03 | *92.70±2.49* | 87.49±11.36 | 87.35±3.60 |
| 5 | 91.50±6.91 | 98.91±1.40 | *99.58±0.40* | 98.65±1.57 | 94.04±0.09 | 93.67±7.30 | 96.85±3.62 | **99.95±0.10** |
| 6 | *86.37±4.16* | 80.49±10.50 | 82.59±5.26 | **86.97±6.39** | 85.88±0.01 | 83.78±7.34 | 82.00±9.34 | 86.28±1.11 |
| 7 | 66.40±12.17 | *85.60±5.57* | 71.40±9.96 | 78.25±9.25 | 77.76±0.12 | 71.91±8.62 | 61.16±9.85 | **90.18±3.31** |
| 8 | 36.51±8.33 | 36.77±15.09 | 47.61±8.37 | *49.28±8.00* | 34.49±0.10 | 46.66±6.18 | 47.30±6.83 | **68.89±2.32** |
| 9 | 61.77±18.69 | 73.30±5.46 | 71.26±2.88 | 80.44±5.96 | *82.87±0.06* | 74.23±7.05 | 60.64±9.57 | **87.29±2.35** |
| 10 | 64.44±16.10 | 57.84±10.05 | 65.59±15.31 | 69.38±19.22 | 61.78±0.37 | *73.62±15.34* | 68.95±12.39 | **84.12±4.58** |
| 11 | 61.61±17.64 | **75.51±13.07** | 65.36±8.46 | 71.59±10.72 | 52.68±0.21 | 64.86±10.54 | 71.20±10.03 | *74.05±6.97* |
| 12 | 59.54±9.55 | 68.50±11.48 | 65.27±15.40 | *75.92±12.54* | 62.03±0.05 | 72.45±11.30 | **77.15±15.93** | 61.70±3.79 |
| 13 | 89.31±6.86 | 76.11±9.36 | 93.38±3.76 | 95.19±2.41 | **99.13±0.00** | 92.39±4.61 | *95.36±1.61* | 68.46±1.53 |
| 14 | 93.07±8.18 | *99.43±0.87* | 93.36±8.30 | 99.10±0.75 | 98.81±0.02 | 96.08±8.34 | 96.50±2.33 | **99.92±0.11** |
| 15 | 96.78±5.37 | *99.13±2.39* | 97.68±2.37 | **99.80±0.40** | 98.35±0.01 | 96.55±4.17 | 98.29±1.97 | 95.11±1.34 |
| OA | 74.23±3.93 | 76.99±1.53 | 78.34±1.67 | *82.68±1.18* | 76.86±1.06 | 79.56±1.83 | 77.81±3.21 | **85.06±0.39** |
| AA | 77.63±3.35 | 79.84±1.35 | 80.95±1.62 | *85.09±1.09* | 81.98±0.57 | 82.13±1.83 | 80.79±2.59 | **85.81±0.40** |
| kappa | 72.16±4.25 | 75.14±1.64 | 76.60±1.80 | *81.29±1.27* | 74.99±1.14 | 77.92±2.00 | 76.03±3.45 | **83.84±0.43** |

mechanisms are limited in the realm of FSHSIC because of significant cross-domain distribution shifts.

### D. Ablation Study

To investigate the contributions of different components in SimHSIC, ablation experiments are conducted. Table VIII lists the classification performance of seven variants of the model in terms of OA, AA, Kappa coefficient, and average change rate (ACR). The ACR represents the average relative change in OA, AA, and Kappa compared with Case 1 (the complete SimHSIC model).

1) Case 1: The proposed SimHSIC.
2) Case 2: SimHSIC with meta-learning, trained on the source domain Chikusei.
3) Case 3: SimHSIC without contrastive learning.
4) Case 4: SimHSIC without contrastive learning but with meta-learning.
5) Case 5: SimHSIC without LPT.
6) Case 6: SimHSIC without LPT but with meta-learning.

To examine the role of meta-learning in SimHSIC, the meta-training procedure from CTF-SSCL is adopted in Cases 2, 4, and 6, with all other settings inherited from Cases 1, 3, and 5, respectively. The ablation results in Table VIII clearly demonstrate the relative importance of each component. Removing LPT (Cases 5 versus 1 and 6 versus 2) leads to the most substantial performance degradation, with ACR drops ranging from −11.79% to −36.80%. Whereas removing contrastive learning (Cases 3 versus 1 and 4 versus 2) results in moderate declines (from −2.11% to −5.91%), indicating that it is helpful but less critical. In contrast, adding meta-learning (Cases 2 versus 1, 4 versus 3, and 6 versus 5) yields inconsistent effects, with ACR changes ranging narrowly from −0.67% to +2.46%. This reflects not only limited improvement but also occasional degradation, suggesting that meta-learning contributes marginally in some scenarios and lacks robustness, demonstrating that its effectiveness is unreliable in practical applications. Overall, these results show that the classification performance of FSHSIC is driven primarily by the LPT, followed by contrastive learning, with meta-learning playing the least significant role.

To further validate the importance of the LPT in FSHSIC, model performance is evaluated under different patch size configurations. Specifically, the performance of CDFS-CASCL, SGPL, and FSCF-SSL is evaluated on four datasets under different patch size configurations, including their respective original sizes ($7 \times 7$, $28 \times 28$, and $33 \times 33$) and a fixed patch size of $9 \times 9$. As shown in Table IX, using larger patch sizes consistently leads to better performance in terms of OA, AA, and Kappa on the four datasets. For example, the OA of FSCF-SSL drops significantly from 75.77% to 60.26% on IP when switching from $33 \times 33$ to $9 \times 9$ patches. Similarly, SGPL performance decreases from 68.19% ($28 \times 28$) to 57.98% ($9 \times 9$) on the same dataset. These results show that the FSHSIC models originally designed with large patch sizes exhibit a significant drop in classification performance when

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TABLE VII
CLASSIFICATION PERFORMANCE (%) ON THE SA DATASET

| Class | FSCF-SSL [8] | CDFS-CASCL [9] | FDFSL [11] | HCFSL-NAS [34] | SGPL [33] | CTF-SSCL [10] | TEFSL [12] | SimHSIC |
|---|---|---|---|---|---|---|---|---|
| 1 | 93.09±9.51 | **99.97±0.06** | 98.96±1.63 | 99.19±1.28 | 94.41±0.06 | 99.54±0.63 | 96.86±0.79 | *99.76±0.24* |
| 2 | 92.68±7.46 | **100.00±0.00** | 98.78±3.38 | *99.94±0.15* | 99.95±0.00 | 99.54±0.93 | 29.96±16.53 | **100.00±0.00** |
| 3 | 85.70±7.80 | *97.22±5.80* | 89.10±1.25 | 96.51±4.00 | 82.77±0.31 | 90.63±11.12 | 48.76±9.05 | **99.97±0.08** |
| 4 | *99.47±0.87* | 97.75±5.77 | 99.32±0.55 | **99.68±0.45** | 99.43±0.00 | 98.58±1.87 | 99.03±0.62 | 98.22±0.41 |
| 5 | 95.79±3.30 | 96.46±2.78 | 90.47±6.25 | 95.10±1.86 | *97.66±0.01* | 91.48±6.77 | 88.28±3.25 | **99.06±0.64** |
| 6 | 99.60±0.20 | *99.85±0.26* | 98.41±2.57 | 99.61±0.65 | 100.00±0.00 | 99.64±0.52 | 99.18±1.19 | **100.00±0.00** |
| 7 | 98.90±1.14 | **99.87±0.22** | *99.76±1.31* | 99.01±1.27 | 97.68±0.01 | 99.64±0.64 | 92.74±12.98 | 99.71±0.18 |
| 8 | 69.11±7.30 | *83.20±4.75* | 78.85±3.97 | 75.21±6.70 | 48.09±0.23 | 78.69±9.08 | 58.24±19.43 | **88.75±1.64** |
| 9 | 97.33±2.42 | 99.90±0.27 | 97.98±1.80 | *99.95±0.11* | 92.05±0.00 | 99.20±0.79 | 98.10±2.20 | **100.00±0.00** |
| 10 | 74.85±13.72 | 92.22±4.84 | 85.40±6.40 | **94.87±1.71** | 84.64±0.06 | 86.55±6.55 | 20.55±5.11 | *94.18±1.07* |
| 11 | 96.07±2.76 | *99.68±0.58* | 95.51±6.65 | 99.66±0.41 | 97.74±0.01 | 97.00±4.17 | 87.90±4.07 | **99.83±0.30** |
| 12 | 97.69±5.03 | **99.76±0.31** | *99.64±0.50* | 99.48±0.69 | 100.00±0.00 | 99.35±0.84 | 89.04±14.65 | 98.61±1.13 |
| 13 | 98.50±1.48 | 98.74±1.39 | 98.29±1.23 | *99.89±0.16* | 99.61±0.00 | 99.64±0.31 | 98.22±0.61 | **99.90±0.06** |
| 14 | 93.70±3.67 | 97.50±4.61 | 99.00±0.93 | *99.09±0.92* | 97.00±0.02 | 98.79±0.84 | 97.75±0.99 | **99.93±0.07** |
| 15 | 46.06±14.77 | **88.95±3.44** | 82.53±6.88 | 73.58±9.02 | 83.73±0.49 | 77.25±9.71 | 48.92±23.63 | *84.95±1.94* |
| 16 | 87.55±8.63 | 95.68±4.40 | 92.04±8.10 | *95.64±5.19* | 85.95±0.14 | 94.17±6.15 | 73.47±3.11 | **99.26±0.21** |
| OA | 82.15±2.20 | *93.95±0.57* | 90.51±1.17 | 90.27±1.29 | 83.46±0.50 | 90.39±1.44 | 69.87±2.81 | **95.08±0.19** |
| AA | 89.13±1.83 | *96.67±0.81* | 93.94±1.26 | 95.40±0.41 | 89.85±0.47 | 94.36±0.92 | 76.69±2.17 | **97.63±0.12** |
| kappa | 80.14±2.46 | *93.27±0.63* | 89.45±1.30 | 89.20±1.42 | 81.74±0.55 | 89.33±1.59 | 66.69±2.97 | **94.52±0.21** |

TABLE VIII
ABLATION RESULTS (%) ON SIMHSIC

| Case | | | | IP | | | | UP | | | | HT | | | | SA | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| No. | LPT | $\mathcal{L}_{con}$ | meta learning | OA | AA | Kappa | ACR | OA | AA | Kappa | ACR | OA | AA | Kappa | ACR | OA | AA | Kappa | ACR |
| 1 | ✓ | ✓ | | *84.88* | *88.74* | *82.91* | — | **91.08** | **89.93** | **88.15** | — | *85.06* | *85.81* | *83.84* | — | **95.08** | **97.63** | **94.52** | — |
| 2 | ✓ | ✓ | ✓ | **86.63** | **92.4** | **84.88** | 2.46 | *90.41* | *89.5* | *87.23* | -0.67 | **87.51** | **87.99** | **86.5** | 2.43 | *95.06* | *97.48* | *94.5* | -0.06 |
| 3 | ✓ | | | 78.95 | 85.41 | 76.67 | -5.17 | 87.42 | 87.13 | 83.45 | -3.72 | 82.81 | 84.59 | 81.42 | -1.96 | 90.01 | 94.41 | 88.84 | -4.66 |
| 4 | ✓ | | ✓ | 80.46 | 87.83 | 77.89 | -3.45 | 88.26 | 87.98 | 84.58 | -2.78 | 83.63 | 85.31 | 82.3 | -1.16 | 91.39 | 95.28 | 90.41 | -3.38 |
| 5 | | ✓ | | 50.88 | 54.12 | 44.27 | **-35.75** | 72.64 | 70.21 | 63.59 | **-20.91** | 71.15 | 73.43 | 68.81 | **-13.77** | 81.2 | 87.57 | 79.1 | **-13.12** |
| 6 | | ✓ | ✓ | 51.98 | 56.41 | 45.12 | *-34.34* | 72.96 | 70.54 | 64.11 | *-20.52* | 75.54 | 76.85 | 73.54 | *-9.59* | 82.69 | 88.29 | 80.69 | *-11.85* |

TABLE IX
EVALUATION OF MODEL PERFORMANCE UNDER ORIGINAL VERSUS UNIFIED PATCH SIZES ($9 \times 9$) ON FOUR DATASETS

| Method | IP | | | | UP | | | | HT | | | | SA | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | OA | AA | Kappa | ACR | OA | AA | Kappa | ACR | OA | AA | Kappa | ACR | OA | AA | Kappa | ACR |
| FSCF-SSL(33*33) | 75.77 | 85.07 | 72.59 | — | 90.55 | 91.54 | 87.63 | — | 78.79 | 81.62 | 77.08 | — | 91.62 | 95.34 | 90.67 | — |
| FSCF-SSL(9*9) | 60.26 | 71.1 | 54.88 | -47.19 | 77.79 | 79.77 | 71.29 | -40.87 | 74.23 | 77.63 | 72.16 | -13.47 | 82.15 | 89.13 | 80.14 | -26.21 |
| CDFS-CASCL(7*7) | 80.98 | 88.77 | 78.52 | — | 83.1 | 90.64 | 78.73 | — | 76.59 | 79.8 | 74.73 | — | 92.97 | 96.26 | 92.18 | — |
| CDFS-CASCL(9*9) | 81.26 | 88.99 | 78.86 | +0.84 | 83.91 | 90.83 | 79.72 | +1.99 | 76.99 | 79.84 | 75.14 | +0.85 | 93.95 | 96.67 | 93.27 | +2.48 |
| SGPL(28*28) | 68.19 | 62.24 | 64.6 | — | 82.58 | 78.07 | 77.09 | — | 75.44 | 77.08 | 73.48 | — | 91.73 | 94.42 | 90.83 | — |
| SGPL(9*9) | 57.98 | 56.82 | 52.93 | -27.3 | 80.21 | 76.99 | 73.67 | -6.87 | 76.86 | 81.98 | 74.99 | +7.83 | 83.46 | 89.85 | 81.74 | -21.93 |

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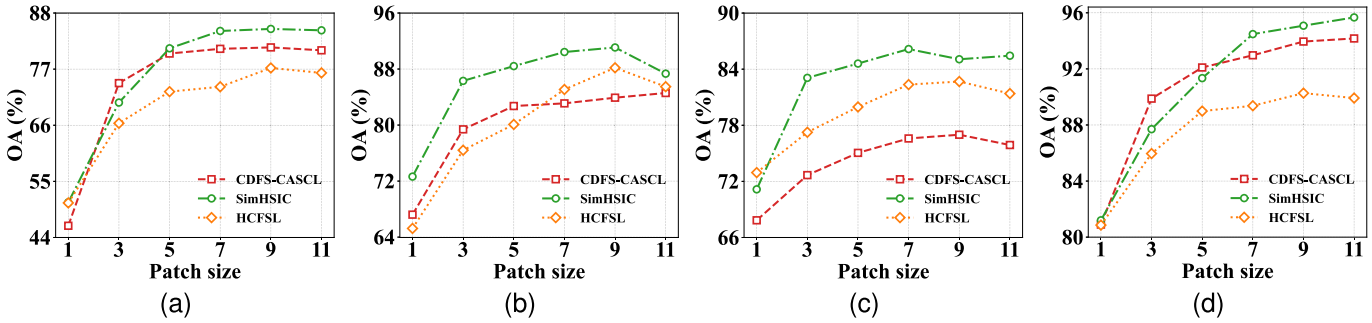



Fig. 4. OA (%) of three representative methods (including two top-performing baselines and SimHSIC) on the four datasets with different patch sizes. (a) IP. (b) UP. (c) HT. (d) SA.

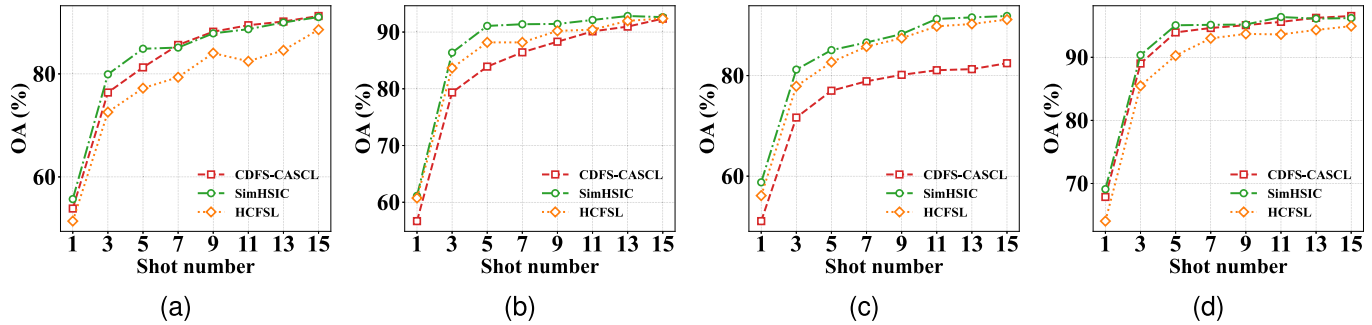



Fig. 5. OA (%) of three representative methods (including two top-performing baselines and SimHSIC) on the four datasets with different shot numbers. (a) IP. (b) UP. (c) HT. (d) SA.

TABLE X
COMPLEXITY ANALYSIS OF DIFFERENT METHODS ON THE UP DATASET

| Metric | Method | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | FSCF-SSL [8] | CDFS-CASCL [9] | FDFSL [11] | HCFSL-NAS [34] | SGPL [33] | CTF-SSCL [10] | TEFSL [12] | SimHSIC |
| Params/K | 210.77 | 311.52 | 334.03 | *190.35* | 307.52 | 390.36 | 959.70 | **105.67** |
| FLOPSs/M | 17.50 | 48.27 | 51.94 | *2.22* | 157.74 | 5.56 | 14.00 | **1.65** |
| GPU memory/MB | 3522 | 4877 | 5262 | *2994* | 4812 | 6145 | 14638 | **1594** |
| Traning time/s | 660.62 | 2093.92 | *555.73* | 2520.43 | 787.40 | 927.32 | 963.74 | **211.80** |
| Test time/s | 4.37 | 3.62 | 8.45 | 150.69 | 8.52 | *2.67* | 12.82 | **1.99** |

smaller patches are used, thereby further demonstrating the critical role of the LPT in FSHSIC.

### E. Parameter Sensitivity

The performance of SimHSIC is largely influenced by two key parameters: the patch size for each sample and the number of shots for each class. The former determines the amount of spatial context available to the model, whereas the latter reflects the density of supervision. Therefore, parameter sensitivity experiments are conducted on four datasets, where the patch size is varied from $1 \times 1$ to $11 \times 11$ and the shot number is varied from 1 to 15. To avoid interaction effects, the shot number is fixed at 5 when the patch size is analyzed, and the patch size is fixed at $9 \times 9$ when the shot number is analyzed. Figs. 4 and 5 show the OA trends under different parameter settings, focusing on SimHSIC and the two most competitive baseline methods—CDFS-CASCL and HCFSL-NAS—according to their top performance in Fig. 2.

1) As shown in Fig. 4, the OA of all three models increases significantly as the patch size increases, highlighting the critical role of the LPT in improving classification performance under few-shot settings. This confirms that larger patches provide richer spatial context, which greatly enhances model discriminability. When the patch size reaches $11 \times 11$, a slight performance drop is observed across several datasets, which can be attributed to the introduction of irrelevant background or interclass interference caused by an overly large receptive field. In addition, SimHSIC achieves the best performance in nearly all cases, which demonstrates its strong classification capability.
2) As shown in Fig. 5, SimHSIC steadily improves OA across all the datasets as the number of shots changes, with smooth and consistent performance trends. On the UP and HT datasets, SimHSIC consistently outperforms the other methods across all shot settings, demonstrating sustained performance advantages. On the SA dataset, SimHSIC achieves the highest accuracy in the low-shot (1–3) and mid-shot (7–11) ranges, while it maintains competitive results at higher shot numbers. Although SimHSIC performs slightly below CDFS-CASCL on the

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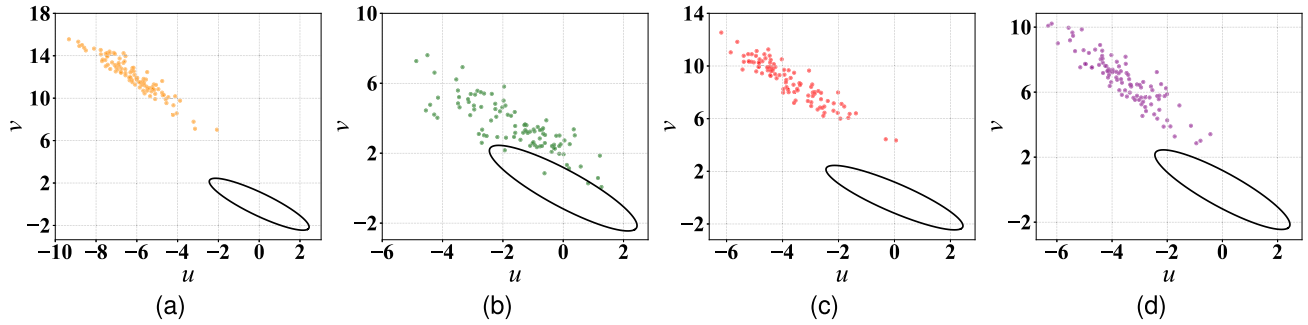



Fig. 6. Visualization of spectral distributional differences between the source domain and three target domains for semantically aligned classes. Each point denotes a Cucconi test statistic $C = f(u, v)$ (the details of $f(u, v)$ can be found in [54]) comparing a source–target sample pair. The elliptical contour represents the 95% confidence region. Points outside the ellipse indicate statistically significant distributional shifts. (a) Grass (Chikusei versus IP). (b) Water (Chikusei versus UP). (c) Asphalt (Chikusei versus HT). (d) Trees (Chikusei versus HT).

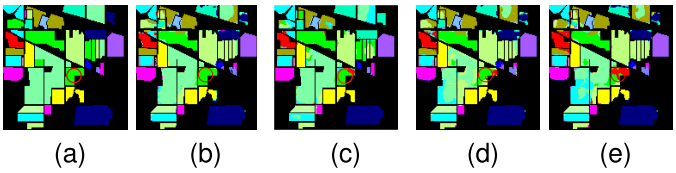



Fig. 7. Classification maps for the IP dataset. (a) Ground truth. (b) SimHSIC. (c) CDFS-CASCL. (d) HCFSL-NAS. (e) FDFSL.

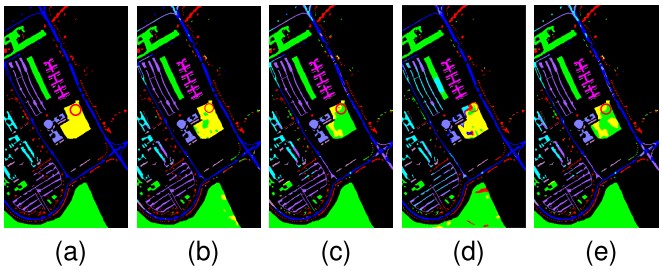



Fig. 8. Classification maps for the UP dataset. (a) Ground truth. (b) SimHSIC. (c) CDFS-CASCL. (d) HCFSL-NAS. (e) TEFSL.

IP dataset from seven shots onward, it remains clearly ahead of HCFSL and retains strong competitiveness. These observations confirm that SimHSIC delivers reliable performance under varying levels of supervision, with particularly notable advantages in low-shot scenarios, validating its practicality and robustness for FSHSIC tasks.

### *F. Computational Complexity Analysis*

As presented in Table X, the computational complexity of all methods is evaluated on the UP dataset in terms of the parameter count, FLOPs, GPU memory consumption, training time, and inference time. SimHSIC has the lowest number of FLOPs (1.65 M), the fewest parameters (105.67 K), and the least GPU memory consumption (1594 MB) among all the compared methods, indicating its lightweight architecture. In terms of efficiency, SimHSIC requires only 211.80 s for training and 1.99 s for inference, both of which are substantially lower than those of advanced meta-learning-based methods.

Compared with state-of-the-art meta-learning baselines, SimHSIC does not rely on source-domain training or complex adaptation stages. Instead, the entire training process is performed using only limited labeled samples from the target domain, resulting in minimal computational overhead.

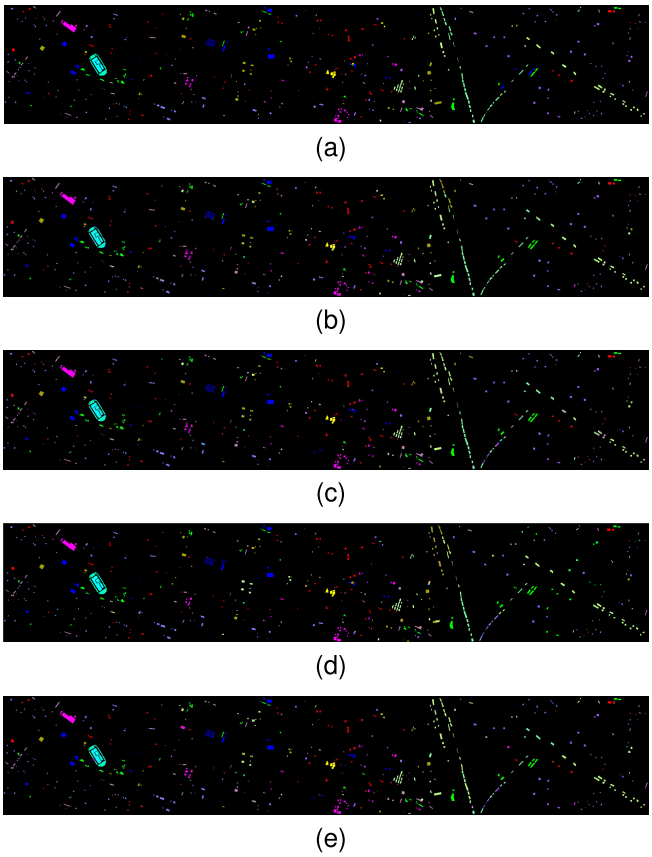



Fig. 9. Classification maps for the HT dataset. (a) Ground truth. (b) SimHSIC. (c) CDFS-CASCL. (d) HCFSL-NAS. (e) CTF-SSCL.

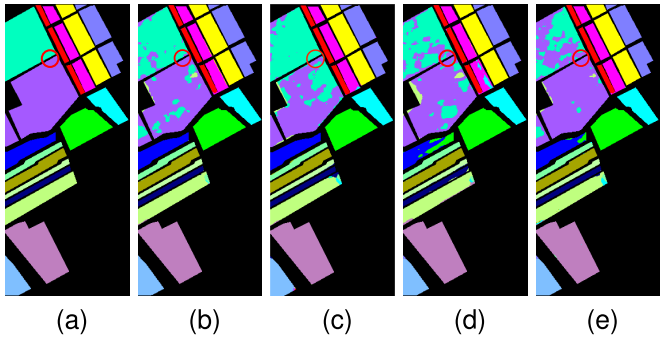



Fig. 10. Classification maps for the SA dataset. (a) Ground truth. (b) SimHSIC. (c) CDFS-CASCL. (d) HCFSL-NAS. (e) FDFSL.

Despite its simplicity, SimHSIC achieves superior classification performance while maintaining extremely low resource

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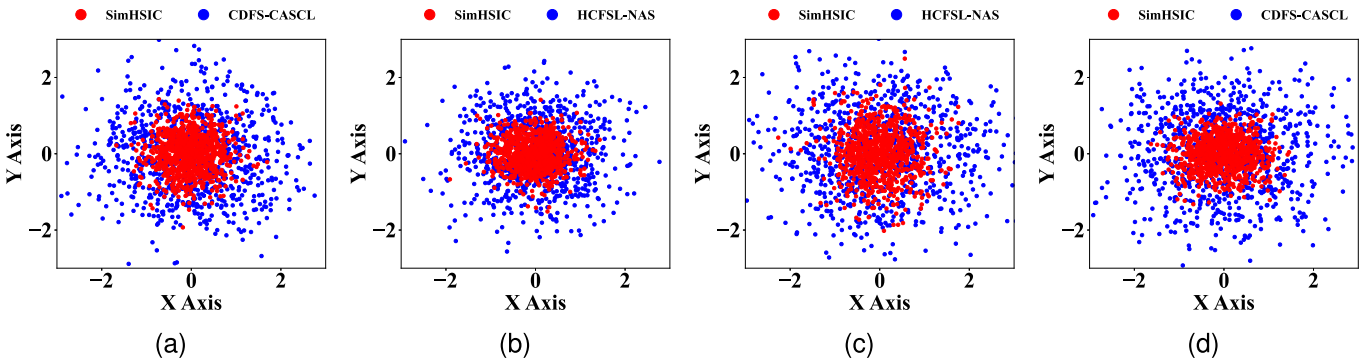



Fig. 11. Visualizations of 2-D t-SNE points of feature distributions of pixels belonging to the same categories generated by SimHSIC and other top-ranking baselines on four datasets. (a) IP. (b) UP. (c) HT. (d) SA. A total of 900 pairs of pixels belonging to the same categories are adopted for each dataset.

consumption, demonstrating that LPT is more important than meta-learning and complicated feature extraction architectures for FSHSIC methods.

### G. Case Study

To further reveal the challenges of spectral alignment, cross-domain distributional discrepancies are analyzed for semantically consistent classes across five widely used hyperspectral remote sensing datasets: Chikusei, IP, UP, HT, and SA. These datasets differ significantly in terms of spatial resolution, spectral dimensionality, and scene types, and they are widely adopted in FSHSIC research. In prior works, Chikusei was typically used as the source domain, whereas IP, UP, and HT were treated as target domains for comparison. Each source–target pair is constructed by randomly selecting 100 semantically aligned samples per class. After downsampling to 100 bands, the spectral data are analyzed via the Cucconi [54] test to evaluate distributional divergence. The results, visualized in Fig. 6, revealed substantial distributional shifts across domains, even for classes with identical semantic labels. This highlights the inconsistency of spectral characteristics under domain variation and refutes the benefits of meta-learning.

To vividly illustrate the visual advantages of SimHSIC in producing cleaner predictions with clearer boundaries and fewer misclassifications, the classification results of four datasets are visualized in Figs. 7–10. It can be observed that SimHSIC results in fewer misclassifications and outperforms all other baselines. Moreover, there are several areas (indicated by red circles in Figs. 7–10) where only SimHSIC is able to make the correct classification. These visualizations support the quantitative findings summarized in Tables IV–VII.

These visual comparisons confirm that SimHSIC provides reliable classification outputs across diverse spatial layouts and scene types. Its predictions exhibit better regional continuity, stronger structural awareness, and higher visual alignment with ground truth, which are critical for real-world FSHSIC tasks. In addition, as shown in Fig. 11, 2-D t-SNE is incorporated to illustrate the feature distributions of pixels belonging to the same categories. The results reveal that the feature maps obtained by SimHSIC form compact clusters, whereas those derived from other top-ranking baselines show relatively scattered distributions. Moreover, statistical significance analysis is conducted by employing the Cucconi rank test to evaluate the pairwise distribution identities of the feature maps corresponding to pixels within the same category. Smaller $C$ statistics indicate more significant pairwise distribution identities. As shown in Table XI, the obtained $C$ statistics demonstrate that SimHSIC achieves the most significant results, indicating that the feature maps produced by SimHSIC are not only numerically superior but also statistically more consistent in distribution compared to those of the baselines. These phenomena indicate that SimHSIC achieves more effective feature alignment, thereby producing more reliable and discriminative representations for classification.

TABLE XI
STATISTICAL COMPARISON OF SIMHSIC AND OTHER TWO TOP-PERFORMING BASELINES BY THE $C$ STATISTICS

| Method | IP | UP | HT | SA |
|---|---|---|---|---|
| SimHSIC | **0.33** | **0.24** | **0.45** | **0.28** |
| CDFS-CASCL | *1.46* | 0.89 | 1.94 | *1.05* |
| HCFSL-NAS | 2.07 | *0.65* | *1.58* | 1.21 |

## VI. CONCLUSION AND FUTURE WORK

This work reveals that the performance advantages of meta-learning-based FSHSIC methods primarily arise from the LPT rather than from the meta-learning paradigm itself. To verify this insight, this work proposes SimHSIC, a lightweight MLP-based baseline that discards meta-learning and prior knowledge entirely. With its minimal architecture and extremely low training cost, SimHSIC achieves state-of-the-art classification accuracy across four benchmarks and outperforms existing meta-learning-based approaches, demonstrating that meta-learning requires further investigation to determine whether it is consistently beneficial and effective for specific FSHSIC scenarios before practical application.

In the future, we will continue to explore FSHSIC and further improve the current SimHSIC to make it more general and practical. The parameter sensitivity analysis showed that the LPT significantly affects classification performance, whereas

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SimHSIC is highly sensitive to patch size, with noticeable performance fluctuations across different settings. Therefore, an intelligent method will be developed to adaptively determine the optimal patch size to enhance model robustness. Enhanced feature expressiveness will be required if more labeled samples are available. In addition, the AA of SimHSIC is slightly lower than that of the top-performing baseline on certain datasets, possibly because of stronger discrimination toward majority classes and weaker recognition of minority classes. To mitigate this imbalance, a dynamic resampling strategy based on class distribution will be explored, with training sample proportions adaptively adjusted to improve minority class discrimination performance.

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